Nonlinear Recurrences, Fixed Points, and Metric-Space Exercises

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§1.1 Finding a fixed point is only the beginning

An iteration takes the form

$$a_{n+1} = f(a_n).$$

If the sequence converges to L and f is continuous at L, then

$$L = \lim_{n\to\infty} a_{n+1} = \lim_{n\to\infty} f(a_n) = f(L).$$

This necessary condition does not guarantee convergence. The map $f(x) = -x$ fixes 0, yet almost every orbit alternates between a_1 and $-a_1$. The map $f(x) = 2x$ also fixes 0, but nonzero orbits move away from it.

For a differentiable map near a fixed point L, the derivative is the first local test. By continuity of f', if $|f'(L)| < 1$ then one can find an interval about L in which f is a contraction, so iterations sufficiently close to L converge geometrically. If $|f'(L)| > 1$, the fixed point is locally repelling. The boundary case $|f'(L)| = 1$ requires more analysis and is particularly relevant to the Möbius example below. This local stability viewpoint is standard in one-dimensional discrete dynamics; the linear-recursion method later in the note follows MIT's *Mathematics for Computer Science*, Chapter 10.[1]

An independent elementary proof method is monotone convergence. If one proves by induction that the iterates stay in a closed interval and that $a_{n+1} \geq a_n$, boundedness gives a limit by completeness of the real numbers. Continuity then identifies the limit with a fixed point. Finding the fixed point without first proving these hypotheses would reverse the logical argument.

§1.2 Linearizing Möbius Recurrences

A fractional-linear map is

$$F(x) = \frac{px + q}{rx + t}, \qquad pt - qr \neq 0.$$

It is associated with the invertible matrix

$$M = \begin{pmatrix} p & q \\ r & t \end{pmatrix}$$

acting on the projective line. Iteration corresponds to the matrix powers M^n up to a common nonzero scalar. This gives an exact algebraic representation of a nonlinear-looking recurrence.

Fixed points satisfy

$$rx^2 + (t - p)x - q = 0.$$

[1] For the recurrence calculation see MIT 6.042J, *Mathematics for Computer Science*, Chapter 10.

When this equation has two distinct finite roots α, β, the coordinate

$$h(x) = \frac{x-\alpha}{x-\beta}$$

maps the fixed points to 0 and ∞. The conjugate map $h \circ F \circ h^{-1}$ fixes both 0 and ∞, so it has the form $u \mapsto \lambda u$. In particular,

$$\frac{a_{n+1}-\alpha}{a_{n+1}-\beta} = \lambda \frac{a_n-\alpha}{a_n-\beta}$$

where the expressions are defined. The new sequence is geometric and its behavior can be read from $|\lambda|$.

Example 1.2.1 (A repeated fixed point reveals a finite-domain obstruction)
The original recurrence is

$$a_1 = \frac{1}{2}, \qquad a_{n+1} = \frac{5a_n - 1}{a_n + 3}.$$

Its fixed-point equation reduces to

$$x^2 - 2x + 1 = (x-1)^2 = 0.$$

The repeated fixed point is 1. The usual quotient $(x-\alpha)/(x-\beta)$ degenerates, but the reciprocal coordinate

$$b_n = \frac{1}{a_n - 1}$$

gives a translation:

$$a_{n+1} - 1 = \frac{4(a_n-1)}{a_n+3}, \qquad b_{n+1} = \frac{a_n+3}{4(a_n-1)} = b_n + \frac{1}{4}.$$

Since $b_1 = -2$, we obtain

$$b_n = -2 + \frac{n-1}{4} = \frac{n-9}{4}, \qquad \boxed{a_n = 1 + \frac{4}{n-9}.}$$

There is an important restriction: the displayed real recurrence is defined only while the denominators are nonzero. In fact,

$$a_8 = -3,$$

so the rule for a_9 requires division by 0. The original real-valued sequence stops at $n = 8$, and it would be incorrect to claim that the closed formula describes an infinite real sequence. On the projective line one can instead interpret $a_9 = \infty$ and continue with $a_{10} = 5$.
The derivative

$$F'(x) = \frac{16}{(x+3)^2}$$

satisfies $F'(1) = 1$. The fixed point is therefore neutral at first order, which is precisely why the reciprocal coordinate and its additive drift are more informative than the elementary contraction test.

The single calculation brings together the two cases: distinct fixed points produce a multiplicative coordinate, while a repeated fixed point produces a translation in a suitable reciprocal coordinate. It also shows why the domain of iteration must be specified before asserting a limit.

§1.3 Repeated Roots of Linear Recurrences

A linear recurrence

$$a_{n+2} = pa_{n+1} + qa_n$$

has characteristic polynomial $P(\lambda) = \lambda^2 - p\lambda - q$. Substituting $a_n = \lambda^n$ verifies why each nonzero characteristic root yields a geometric solution. If the two roots λ_1, λ_2 are distinct, every solution has the form

$$a_n = A\lambda_1^n + B\lambda_2^n,$$

with constants determined by two initial conditions.

For a repeated root $\lambda \neq 0$, a second independent solution is $n\lambda^n$, yielding

$$a_n = (A + Bn)\lambda^n.$$

The factor n is the discrete counterpart of a generalized eigenvector: the companion matrix

$$\begin{pmatrix} p & q \\ 1 & 0 \end{pmatrix}$$

has a Jordan block when the characteristic root repeats. Thus repeated roots are not merely an exception in a formula; they are a test of whether diagonalization succeeds.

The matrix picture also explains the Möbius case. Fractional-linear iteration uses a projective 2×2 matrix, while a second-order linear recurrence uses a companion 2×2 matrix. Both problems are governed by eigenvalues and invariant directions, but the transformation from a vector to its ratio changes the representation of the dynamics.

§1.4 Two Constructions of Spaces

The original material also includes problems about Dedekind cuts and metric subspaces. These concern *different* questions from recurrence linearization and deserve their own mathematical explanation rather than being forced into the recurrence story.

For a Dedekind cut $X \subset \mathbb{Q}$, use the usual convention that X is a nonempty proper downward-closed set without a greatest element. For two cuts X, Y, define

$$X + Y = \{x + y : x \in X,\ y \in Y\}.$$

Why is this again a cut? It is nonempty by choosing $x \in X, y \in Y$. It is downward closed: if $q < x + y$, then $q = (q - y) + y$ with $q - y < x$, hence $q - y \in X$. It has no greatest element because X has none: for $x + y$, choose $x' > x$ still in X. Finally, choose rational upper bounds B_X, B_Y of X, Y; then every element of $X + Y$ is less than $B_X + B_Y$, proving it is proper. This checks the four cut axioms explicitly.

For a metric space (X, d) and a subset $Y \subseteq X$, the restriction

$$d_Y(y_1, y_2) = d(y_1, y_2)$$

is again a metric, since each metric axiom holds for all points in X, hence in Y. In Euclidean space,

$$d(x, y) = \sqrt{\sum_{i=1}^{n} (x_i - y_i)^2},$$

and the triangle inequality follows from the inner-product Cauchy–Schwarz inequality:

$$\|u + v\|_2^2 = \|u\|_2^2 + 2\langle u, v\rangle + \|v\|_2^2 \leq (\|u\|_2 + \|v\|_2)^2.$$

These exercises establish how an existing order or distance structure can be extended or restricted, without pretending that the resulting objects are instances of the same iteration theorem.